IMU Sensor Simulation and Complementary Filter Based Sensor Fusion in MATLAB

A Single-Axis Modeling and Performance Evaluation Study

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This report presents a MATLAB-based simulation study of single-axis IMU behavior and complementary filter based sensor fusion.

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Abstract

Inertial Measurement Units (IMUs) are widely used for orientation and motion estimation in aerospace and embedded systems. However, individual IMU sensors suffer from inherent limitations. Gyroscopes provide smooth angular velocity measurements but are affected by bias drift, leading to accumulated angular error over time. Accelerometers estimate tilt using gravity but are highly sensitive to measurement noise.

This project presents a MATLAB-based single-axis IMU simulation that models gyroscope bias, time-varying drift, and Gaussian noise, along with accelerometer noise characteristics. A complementary filter is implemented to fuse both sensor outputs and mitigate their individual weaknesses.

Performance evaluation using Root Mean Square Error (RMSE) demonstrates significant improvement in estimation accuracy through sensor fusion. Frequency-domain analysis using Fast Fourier Transform (FFT) further validates the reduction of high-frequency noise while preserving the dominant motion frequency.

Keywords: IMU, Sensor Fusion, Complementary Filter, Bias Drift, MATLAB Simulation, RMSE, FFT

1 Introduction

1.1 Background

Inertial Measurement Units (IMUs) are essential components in modern navigation, robotics, aerospace systems, and embedded motion tracking applications. A typical IMU consists of a gyroscope and an accelerometer, which provide angular velocity and linear acceleration measurements respectively. These measurements can be used to estimate the orientation of a rotating body.

Accurate orientation estimation is critical in applications such as aircraft stabilization, drone navigation, satellite attitude control, and robotic motion analysis. However, raw sensor measurements alone are insufficient for reliable long-term estimation.

1.2 Problem Statement

Despite their widespread use, individual IMU sensors exhibit inherent limitations. Gyroscopes provide smooth short-term angular velocity measurements but suffer from bias drift. Even a small bias, when integrated over time, results in an accumulated angular error. This leads to a long-term instability in angle estimation.

Accelerometers, on the contrary, estimate tilt using the gravity vector but are highly sensitive to measurement noise and external disturbances. As a result, accelerometer-based estimation appears noisy and unstable in the short term.

Relying on either sensor independently leads to an inaccurate orientation estimation.

1.3 Objective

The objective of this project is to develop a single-axis IMU simulation in MATLAB that models gyroscope bias, time-varying drift, and accelerometer noise characteristics. A complementary filter is implemented to fuse both sensor outputs in order to reduce estimation error.

System performance is evaluated quantitatively using the root mean square error (RMSE) and validated through frequency-domain analysis using the fast Fourier Transform (FFT).

2 System Modeling

2.1 True Motion Model

To simulate controlled rotational motion, a sinusoidal angle profile is defined as the ground truth reference:

$$\theta(t) = A\sin(2\pi f t)$$

For this study, the amplitude is chosen as $A = 20°$ and the motion frequency as $f = 0.2$ Hz, resulting in:

$$\theta(t) = 20\sin(2\pi \cdot 0.2\,t)$$

This represents one full oscillation every 5 seconds. The signal is sampled at 100 Hz to mimic realistic IMU conditions and provide adequate resolution for integration and frequency analysis. It serves as the reference benchmark for evaluating all sensor estimates.

2.2 Gyroscope Model

The gyroscope measures angular velocity, which is the time derivative of the true angle. Differentiating the sinusoidal motion gives:

$$\omega_{true}(t) = \frac{d\theta(t)}{dt} = 20 \cdot 2\pi \cdot 0.2\cos(2\pi \cdot 0.2\,t)$$

In practical systems, gyroscope measurements are affected by bias and noise. The simulated gyroscope output is modeled as:

$$\omega_{measured}(t) = \omega_{true}(t) + b(t) + n(t)$$

where $b(t)$ represents time-varying bias and $n(t)$ represents zero-mean Gaussian noise.

When angular velocity is numerically integrated to estimate angle, bias introduces cumulative drift:

$$\theta_{error}(t) = \int b(t)\,dt$$

Thus, even small bias values result in increasing estimation error over time.

2.3 Accelerometer Model

The accelerometer estimates tilt angle using the gravity vector. In ideal static conditions, tilt can be computed from gravitational components.

In this simulation, accelerometer output is modeled directly as:

$$\theta_{acc}(t) = \theta_{true}(t) + n_{acc}(t)$$

where $n_{acc}(t)$ represents zero-mean Gaussian noise.

Unlike gyroscope-based estimation, accelerometer measurements do not accumulate drift over time. However, they exhibit high-frequency noise, leading to short-term instability in angle estimation.

3 Complementary Filter Based Sensor Fusion

3.1 Fusion Principle

The gyroscope provides accurate short-term angular rate information but suffers from long-term drift due to bias. The accelerometer provides stable long-term tilt reference but exhibits high-frequency noise.

To combine the advantages of both sensors, a complementary filter is implemented. The complementary filter effectively performs a high-pass operation on the gyroscope signal and a low-pass operation on the accelerometer signal.

3.2 Filter Equation

The discrete-time complementary filter is expressed as:

$$\theta_k = \alpha\left(\theta_{k-1} + \omega_k \Delta t\right) + (1-\alpha)\theta_{acc,k}$$

where:

- θ_k is the fused angle estimate
- ω_k is the measured angular velocity
- $\Delta t = \frac{1}{100}$ seconds is the sampling interval
- $\theta_{acc,k}$ is the accelerometer-based angle
- α is the filter coefficient

In this study, $\alpha = 0.98$ is selected to balance short-term gyroscope stability and long-term accelerometer correction. A high α value ensures smooth angular integration from the gyroscope while allowing gradual drift compensation from the accelerometer. This choice provides a favorable trade-off between noise sensitivity and bias-induced drift, as validated by the lowest RMSE observed in the results.

MATLAB Implementation Snippet

The complementary filter is implemented in MATLAB as:

```
alpha = 0.98;
complementary_angle = zeros(size(t));
complementary_angle(1) = theta_true(1);

for k = 2:length(t)
    gyro_integration = complementary_angle(k-1) + gyro_output(k)*dt;
    complementary_angle(k) = alpha * gyro_integration + ...
                             (1-alpha) * accel_output(k);
end
```

3.3 Interpretation

The term $(\theta_{k-1} + \omega_k \Delta t)$ represents gyroscope-based integration, which captures smooth short-term motion. The term $(1-\alpha)\theta_{acc,k}$ introduces a small correction factor that prevents long-term drift.

By tuning α, a trade-off is established between noise sensitivity and drift correction. Higher values of α increase reliance on gyroscope data, while lower values increase accelerometer influence.

4 Results and Performance Analysis

This section shows the time-domain performance of the simulated IMU system, comparing individual sensor estimates with the complementary filter to assess drift, noise, and overall accuracy.

4.1 Gyroscope-Only Estimation

Figure 1 shows the angle obtained by integrating the gyroscope output. Due to time-varying bias, the estimate gradually drifts from the true angle over time.

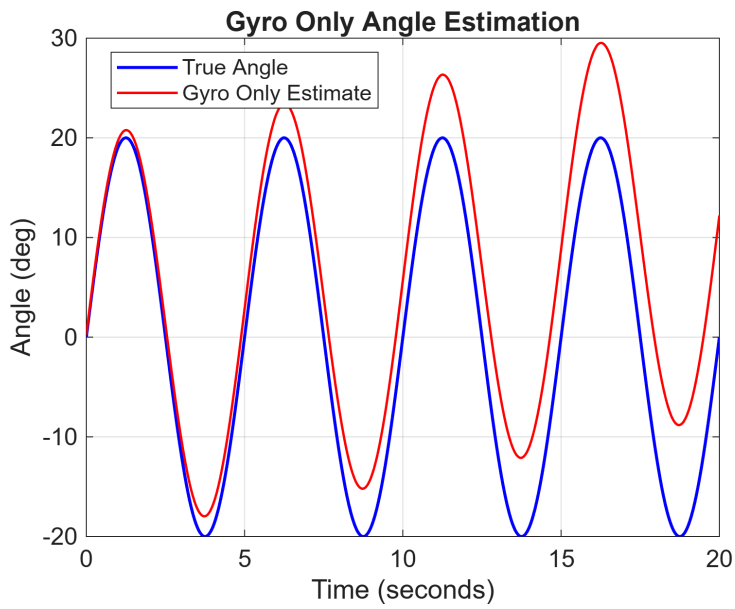

Figure 1: Comparison of True Angle and Gyroscope-Only Estimation

4.2 Accelerometer-Only Estimation

Figure 2 presents the angle estimated directly from the accelerometer measurement. Unlike gyroscope integration, the accelerometer does not exhibit cumulative drift. However, significant high-frequency noise is observed in the signal.

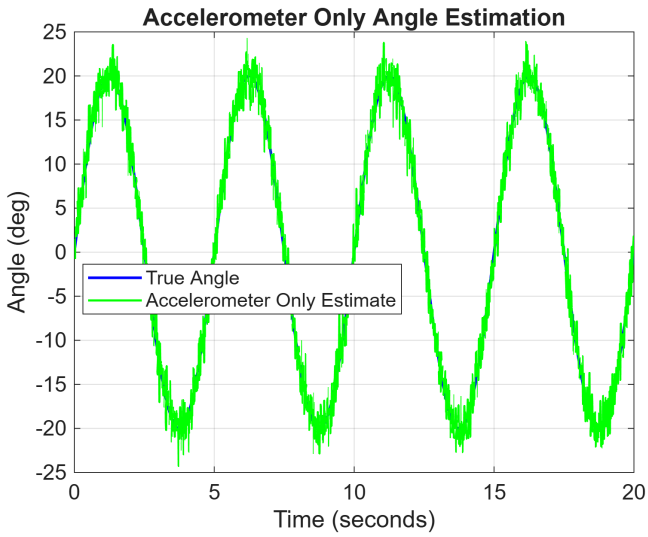

Figure 2: Comparison of True Angle and Accelerometer-Only Estimation

4.3 Complementary Filter Estimation

Figure 3 shows the fused angle obtained using the complementary filter. The output closely follows the true angle while significantly reducing both long-term drift and high-frequency noise.

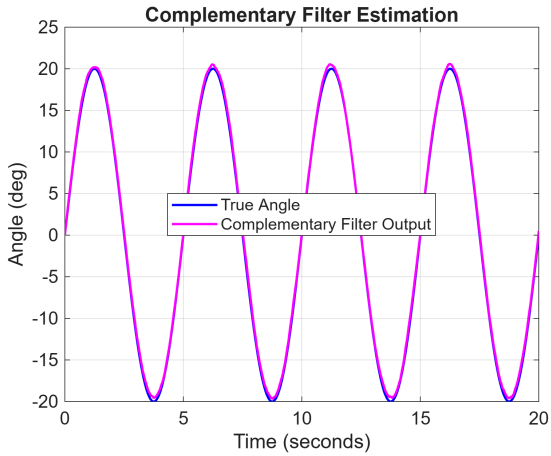

Figure 3: Comparison of True Angle and Complementary Filter Estimation

4.4 Quantitative Error Analysis

To objectively evaluate performance, the Root Mean Square Error (RMSE) is computed between the true angle and each estimated signal.

Method	RMSE (degrees)
Gyroscope Only	6.767
Accelerometer Only	1.514
Complementary Filter	0.314

The complementary filter has the lowest RMSE, reducing gyro drift and accelerometer noise, confirming better accuracy than standalone sensors.

4.5 Frequency-Domain Analysis

To further validate the filtering performance, Fast Fourier Transform (FFT) analysis is performed on the estimated signals.

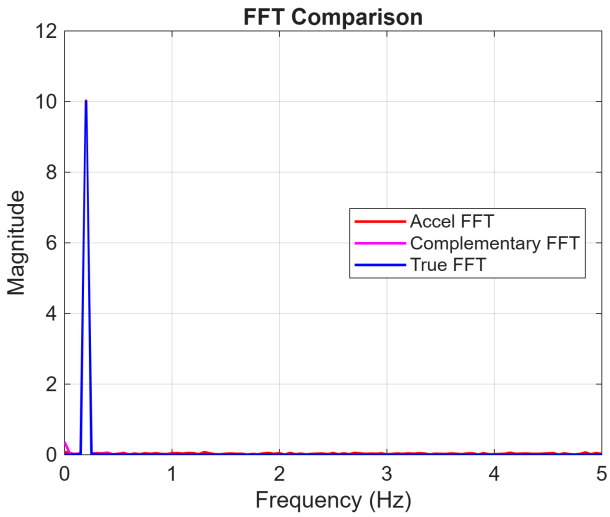

Figure 4: FFT Comparison of Accelerometer and Complementary Filter Outputs

The dominant spectral peak at 0.2 Hz corresponds to the simulated motion frequency. While the accelerometer signal exhibits significant high-frequency components due to noise, the complementary filter preserves the fundamental motion frequency while attenuating higher-frequency noise components.

This confirms that the complementary filter effectively combines low-frequency stability from the accelerometer with high-frequency smoothness from the gyroscope.

5 Conclusion and Future Scope

This project presented a single-axis IMU simulation developed in MATLAB to analyze the effects of gyroscope bias drift and accelerometer noise on angle estimation. The complementary filter was implemented to fuse both sensor measurements and mitigate their individual limitations. Time-domain results showed that gyroscope-only estimation suffers from cumulative drift, while accelerometer-only estimation exhibits high-frequency noise. The complementary filter effectively mitigated both effects, closely tracking the true angle. RMSE analysis confirmed substantial accuracy improvement, and frequency-domain evaluation demonstrated preservation of the fundamental motion frequency with reduced noise components.

Overall, this study demonstrates the effectiveness of complementary filtering for tilt estimation and provides a clear foundation for extending the approach to multi-axis IMU systems.

Future Scope

Future improvements may include extending the model to three-axis orientation estimation, implementing a Kalman filter for optimal state estimation, and validating the algorithm using real IMU hardware data.